

CLIP-KO: Knocking Out Typographic Attacks in Contrastive Language-Image Pre-Training

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Abstract

Recent advances in Vision Transformers (ViTs), particularly within the CLIP family of models, have brought unprecedented capabilities in zero-shot vision-language understanding. However, persistent vulnerabilities remain, most notably typographic attacks that exploit the model’s “text reading obsession” (typographic attack vulnerability [9]) by injecting spurious or adversarial text into images. We hypothesize that these attacks are tied to the emergence of high-norm “register tokens” [7] and the unique role of specific “register neurons” [11] and “global accumulation” attention heads (see Figures 1 and 2) within the ViT architecture. In this work, we present **CLIP-KO**: a robust fine-tuning framework for OpenAI’s CLIP ViT-L/14 that combines key projection orthogonalization loss, attention head dropout, geometric parametrization, and adversarial training. Our approach dramatically improves typographic attack resilience and enhances performance across a suite of vision-language benchmarks; all with no architectural changes and using only modest compute (single RTX 4090 GPU). We provide an overview mechanistic analysis of register neurons and attention heads, demonstrate invariance of the register mechanism. We release our code, models, and adversarial datasets for reproducibility (<https://github.com/zer0int/CLIP-fine-tune>, <https://huggingface.co/zer0int>).

1 Introduction

Contrastive Language-Image Pre-training (CLIP) has established itself as a de-facto standard for learning transferable vision-language representations, enabling strong zero-shot classification and retrieval performance across diverse datasets. Despite its widespread adoption, CLIP and related ViT-based models exhibit persistent vulnerabilities to adversarial / typographic attacks. Visible text in images - be it natural or synthetic (inserted) - induces misclassifications by exploiting internal salience to text features in images [9, 1].

The mechanistic roots of Vision Transformers are increasingly well-understood. Recent work has identified that ViT models re-purpose certain patch tokens as “register tokens”: High-norm outliers in the sequence accumulate global context for internal computation [8]. Our own empirical studies reveal that a small, stable set of a total of 13 “register neurons” in Layer 11 and Layer 12 of the CLIP ViT-L-14 model are tied to a spike in L2 output norm of a specific attention head (Head 10) (see Figures 1). While ablation or reallocation of these neurons and tokens has been shown to reduce attention heatmap artifacts [10], such approaches don’t address the underlying issues (e.g. typographic attack vulnerability).

Building on these findings, we propose a robust fine-tuning framework for CLIP ViT-L/14, **CLIP-KO**, that preserves the beneficial internal mechanisms of register tokens while mitigating their role in typographic attack vulnerability and attention heatmap artifacts. Our approach combines several key innovations:

- **Key Projection Orthogonalization:** Encourages decorrelation among attention heads, limiting redundancy and spurious co-adaptation.
- **Attention Head Dropout:** Randomly ablates entire attention heads during training to improve generalization and robustness.
- **Geometric Parametrization (GmP):** Applies the GmP technique [6] to all MLP layers, decoupling radial and angular weight components for improved optimization and regularization.
- **Adversarial Training:** Integrates a targeted adversarial dataset, collected using SAE-informed embedding [16], to directly fine-tune the model’s response to typographic attacks.

We empirically demonstrate that these interventions, while minimal in computational and architectural overhead, result in dramatic improvements in typographic attack resilience over previously described outcomes [14]. We further observe and strong gains across a comprehensive suite of benchmarks (see Appendix Table 1) including zero-shot, retrieval, and linear probe. Our analysis leverages diagnostic tools - such as per-head norm plots, ablation studies, attention visualization, and key norm visualizations - to elucidate the mechanistic effects of our approach.

Our models, code, and adversarial datasets are released under an open license (MIT) to support further research, benchmarking, and deployment.

2 Methods

We fine-tuned the OpenAI CLIP ViT-L/14 model using the original CLIP codebase (<https://github.com/openai/CLIP>), with the following modifications: To expose the attention QKV and attention weights, we build on previous work for CLIP interpretability by Chefer et al. [5]; for and only during fine-tuning, we apply (1) Geometric Parametrization to all MLP layers, and (2) Attention Head Dropout in the transformer self-attention blocks.

2.1 Geometric Parametrization

To improve robustness and regularization of the transformer MLPs, and to prevent overfit during fine-tuning with small batch sizes, we replace standard linear layers with *geometric parametrization* [6]. In this approach, each weight matrix is factorized into a learnable radial component and a normalized angular component:

$$\mathbf{W}_{\text{geo}} = \mathbf{r} \cdot \mathbf{u}, \quad \text{where } \mathbf{u} = \frac{\theta}{\|\theta\|_2}$$

Here, $\mathbf{r}$ is a learned scaling vector, and θ is a weight matrix whose rows are normalized to unit ℓ_2 norm. This parametrization decouples weight length and direction, constraining the expressiveness of each neuron. Geometric parametrization was applied to both the feedforward expansion and projection layers within each residual block.

2.2 Attention Head Dropout

Inspired by regularization for attention heads, DeCAtt [3], we introduce *Attention Head Dropout*: at each forward pass, attention heads are independently dropped with probability p (default $p = 0.2$), zeroing their output and rescaling the remaining heads to preserve output norm. For attention output $\mathbf{A} \in R^{B \times H \times T \times D}$,

$$\mathbf{A}' = \frac{\mathbf{A} \odot \text{Mask}}{1 - p}$$

where $\text{Mask} \in \{0, 1\}^{B \times H \times 1 \times 1}$ is sampled per head, per sample. This is functionally analogous to Dropout at the head level. Effects observed:

- Improved generalization by reducing head co-adaptation;
- Increased decorrelation/specialization among heads;
- No notable training instability.

2.3 Loss Functions

Three loss terms were used during fine-tuning:

1. Contrastive Loss. We use the standard CLIP contrastive objective as the primary loss, as described in the OpenAI / CLIP paper [13].

2. Key Projection Orthogonality Loss (k-proj ortho). To further decorrelate attention heads, we introduce an orthogonality loss on key projection matrices for selected layers (blocks) in the center of the transformer, covering blocks where register neurons first emerge (as evident via a sudden, dramatic increase in maximum patch norms associated with the emergence of 'register tokens'). For each layer, the key projection is reshaped into head slices, and cosine similarity between all head pairs (per feature dimension) is penalized:

$$\mathcal{L}_{\text{ortho}} = \frac{1}{N_{\text{layers}}} \sum_{\ell} \frac{1}{H(H-1)F} \sum_{i \neq j} \sum_f \left(\left\langle W_i^{(\ell)}[:, f], W_j^{(\ell)}[:, f] \right\rangle \right)^2$$

where $W_i^{(\ell)}$ is the normalized key projection for head i in layer ℓ , H is the number of heads, and F the expanded feature dimension. The **k-proj ortho** loss was developed by us, based on empirical analysis of head redundancy.

3. DeCAtt Loss. The DeCAtt loss [3], in addition, penalizes correlation between per-head attention outputs within a block. We compute the off-diagonal correlation of per-head outputs (after attention, before merging). This loss was always near zero in practice, acting only as a safety constraint.

Loss Composition. All losses are combined with empirically chosen weights, with contrastive loss always dominant; orthogonality and DeCAtt losses act as regularizers.

2.4 Datasets and Adversarial Training Integration

COCO-SPRIGHT-40k. Our fine-tuned models use the COCO-SPRIGHT-40k dataset [4], which includes around 40,000 images (train) with 'spatially right' (SPRIGHT) LLM-generated captions. This dataset, together with geometric parametrization, has shown strong performance in spatially grounded models [15]. Dataset link: https://huggingface.co/datasets/SPRIGHT-T2I/spright_coco.

Adversarial Typographic Attack Dataset. To test and improve adversarial robustness, we construct a dataset of challenging adversarial images with equally challenging (biased) labels. Final set: <https://huggingface.co/datasets/zer0int/CLIP-K0-Adversarial-Train-Typo-Attack>.

Relabeling of Adversarial Set. A subset of 236 images was relabeled with three alternative text labels per image:

1. **CLIP-Driven Label:** The model-predicted label maximizing or minimizing cosine similarity (via CLIP text embedding gradient ascent).
2. **Maximum-Bias Human Label:** A label reflecting strongest bias in the CLIP model, written by the author.
3. **Ground Truth:** The correct, human-annotated label, also written by the author.

Adversarial Loss Integration. During fine-tuning, adversarial image/text triplets were injected every 5 batches (adversarial batch size 1). For each, all three candidate labels were paired with the image, and the adversarial loss computed as:

$$\mathcal{L}_{\text{adv}} = \text{sim}(\text{img}, t_0) + \text{sim}(\text{img}, t_1) + (1 - \text{sim}(\text{img}, t_2))$$

where t_0 and t_1 are the **CLIP-Driven Label** and **Maximum-Bias Human Label**, t_2 is the **Ground Truth** label. This incentivizes correct classification while penalizing agreement with adversarial, typographic, or heavily biased attractors. The adversarial loss is linearly combined with the main loss.

Full Training Loss. The total loss for parameter updates is:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{contrastive}} + \lambda_{\text{ortho}}\mathcal{L}_{\text{ortho}} + \lambda_{\text{decatt}}\mathcal{L}_{\text{decatt}} + \lambda_{\text{adv}}\mathcal{L}_{\text{adv}}$$

with λ coefficients tuned on a held-out validation set.

2.5 Training Details

For all experiments, we fine-tuned the model for 20 Epochs using a batch size of 18 (main) and 1 (adversarial), with adversarial samples injected every 5 steps. Gradient accumulation was set to 2 (effective batch size 36 for main loss). Learning rates: 3×10^{-7} initial, 5×10^{-7} max, with OneCycleLR scheduler (cosine annealing, 30% ramp-up). AdaBelief optimizer: $\epsilon = 1\text{e-}14$, $\beta = (0.9, 0.999)$, weight decay 1e-3, decoupled, rectified. Parameter groups received differentiated learning rates:

- Visual transformer: 3×10^{-7}
- Text transformer: 1×10^{-8}

- Token embeddings: 3×10^{-7}
- Positional/class embeddings: 1×10^{-7}
- Projection heads: 1×10^{-7}
- LayerNorm/convolutional weights: 1×10^{-7}

All training was performed on a single NVIDIA RTX 4090 GPU (total wall-clock ~ 10 hours), illustrating that compute-efficient fine-tuning of large models is feasible with careful optimization and robust methods.

3 Results

3.1 Model Variants and Robustness Tradeoffs

We present and release two main model variants:

- **KO-CLIP:** Fine-tuned with adversarial labels selected (via gradient ascent) as CLIP’s own “max cosine similarity” tokens.
- **KO-LITE:** Fine-tuned identically, but with “min cosine similarity” (“not-image direction”) tokens for adversarial training.

Both leverage k-proj-orthogonality loss, DeCAtt loss, Geometric Parametrization (GmP), and Attention Head Dropout. For ablation, a model trained with special losses but *without* adversarial dataset exposure showed only modest gains regarding typographic attack vulnerabilities (albeit higher gains than our model trained with the adversarial dataset alone and without k-proj ortho loss and head dropout), confirming adversarial training is an essential component for typographic attack robustness.

Out-of-Distribution Embedding Effects. KO-CLIP demonstrates superior typographic attack resilience and outperforms pretrained CLIP by a large margin (SCAM benchmark, zero-shot accuracy: 0.4165 -j 0.7823). However, when used as a text encoder for guiding Flux.1-dev (a generative text-to-image DiT model [2]), KO-CLIP embeddings appear OOD for adversarial or “non-word” (non-standard English language) prompts (see Figure 6). We cannot guarantee that this eliminates all prompt injection risks, nor that the model is robustly SFW-aligned when ‘gibberish words’ are used. Nonetheless, this is a promising direction for defense against prompt injection attacks not detectable by surface NLP methods or blacklisting. However, it also comes at the cost of the model’s ‘creativity’. As we believe that a careful choice for the benefits and downsides of each model’s deployment should be made on a case-by-case basis, it is not our intent to overly restrict models from being able to generate legitimate (including adult) entertainment. Our ideal model should be resilient to typographic attacks while also being otherwise unconstrained for correct classifications. However, we consider inferring adult themes for an unambiguous image of a street sign that says ”DATE STREET” a clear misclassification that should be mitigated; see: https://huggingface.co/datasets/zer0int/CLIP-adversarial-typographic-attack_text-image. We also propose KO-LITE:

KO-LITE: Modality Gap and Benchmark Tradeoffs. KO-LITE (min-cos-sim labels) achieves nearly the same robustness to typographic attacks as KO-CLIP, while exceeding KO-CLIP performance on some other benchmarks (such as linear probe), while being able to understand ‘prompt injections’ and creative ‘CLIP words’ to the same extent as the pre-trained model.

3.2 Adversarial and Benchmark Results

On a canonical “worst-case” typographic attack—ImageNet “goldfinch” images with the word “bumblebee” overlaid (chosen for color similarity and high salience):

- **Pre-trained CLIP:** 2 correct, 58 misclassified.
- **Ours, no adversarial training:** 27 correct, 33 misclassified.
- **Ours, “min-cos-sim” adversarial:** 53 correct, 7 misclassified.
- **Ours, “max-cos-sim” adversarial:** 59 correct, 1 misclassified.

Across broader benchmarks (*see Appendix Table 1 for the complete results*), we observe:

- **BLISS / SCAM typographic attack:** SCAM: Pre-trained 0.4165, KO-CLIP 0.7823, KO-LITE 0.7367.
- **BLISS / SynthSCAM typographic attack:** SynthSCAM: Pre-trained 0.3219, KO-CLIP 0.7358, KO-LITE 0.6790.
- **ImageNet 1k zero-shot (LAION/CLIP_Benchmark):** KO-LITE achieves acc1/acc5 of 0.4688/0.6885, KO-CLIP 0.4344/0.6530, both much better than pre-trained (0.3269/0.5299).
- **VOC 2007 multilabel zero-shot (mAP):** KO-CLIP: 0.8579, KO-LITE: 0.8626, pre-trained: 0.7615.
- **Zero-shot retrieval (MSCOCO, xm3600):** Recall@5 increases from 0.2196 to 0.3296/0.3385 (images) and from 0.3032 to 0.4396/0.4745 (text).

For full results, detailed metric definitions, and more benchmarks, see Appendix Table 1.

Conclusion: Both adversarially fine-tuned models—especially with “max-cos-sim” labels—achieve state-of-the-art typographic robustness and strong performance across a broad suite of vision-language benchmarks, with tradeoffs discussed above. All models and code, including evaluation code, are released for research and reproducibility.

Appendix: Full Results Table

4 Discussion

Our findings substantiate the central role of register neurons and specific attention heads in mediating both the strengths and vulnerabilities of CLIP ViT models. Notably, these mechanisms exhibit remarkable invariance: regardless of fine-tuning regime, the same register neuron indices and head 10 activity persist. This architectural property offers both opportunities for targeted intervention and potential challenges for broader generalization.

Our experiments demonstrate that robust fine-tuning—incorporating geometric parametrization, orthogonalization, attention head dropout, and adversarial training—can significantly mitigate typographic attack vulnerability while maintaining or improving generalization across benchmarks. Importantly, our approach achieves these gains without requiring architectural changes or reliance on proprietary data or compute resources, thus democratizing robust CLIP fine-tuning.

However, several limitations remain. Large-scale evaluation are computationally infeasible within the scope of this work. Claims of adversarial prompt resistance are thus limited to the tested

scenarios; broader generalization requires community benchmarking and resource sharing. Additionally, while our adversarial dataset and code are released for transparency and reproducibility, we caution that dual-use risks remain: the adversarial fine-tuning strategies employed here could be inverted to amplify, rather than mitigate, undesirable behaviors or biases, as outlined in the original CLIP model card [12].

We encourage the community to build on these findings, and to evaluate the generalizability of adversarial robustness strategies across model architectures and domains.

Disclosure

This paper was written with the help of GPT-4.1. I, the human author, provided the rough framework, describing my findings, and answering interview-style questions the AI posed to me upon being prompted to do so. In addition, I have diligently reviewed the paper for correctness to the best of my ability. The summaries and LaTeX were generated (translated) by GPT-4.1 (from my human language input and/or code input), and may alas contain mistakes (“hallucinations”) or otherwise not follow a rigorous scientific standard. As I, the human, am the sole author and doing this independently and in my free time, I’m rooting for your understanding. Criticism and corrections or requests for collaboration are highly welcome; please do reach out (by eMail, or by opening an Issue on GitHub)! This paper is intended to be a summary, an overview. I cannot guarantee completeness or correctness in all aspects. Instead of taking the claims made here at face value, I invite you to let my CLIP model and code speak for itself.

Figures

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Table 1: Full quantitative results for all evaluated benchmarks. $\uparrow$ indicates higher is better; $\downarrow$ lower is better. KO-CLIP: max-cos-sim labels; KO-LITE: min-cos-sim labels.

Benchmark / Metric	Pre-Trained	KO-CLIP	KO-LITE
RTA 100 Typographic Attack			
Zero-Shot Accuracy	0.4330	0.7210	0.6260
BLISS / SCAM Typographic Attack			
NoSCAM (clean)	0.9905	0.9897	0.9897
SCAM	0.4165	0.7823	0.7367
SynthSCAM	0.3219	0.7358	0.6790
ILSVRC2012 Linear Probe			
Top-1 (%)	69.86	70.58	72.65
Top-5 (%)	92.70	93.79	94.08
MVT ImageNet/ObjectNet (zero-shot)			
Accuracy	0.846	0.898	0.9029
ImageNet 1k, Zero-Shot (LAION/CLIP_Benchmark)			
Acc@1	0.32696	0.43440	0.46882
Acc@5	0.52997	0.65297	0.68845
Mean Per-Class Recall	0.32609	0.43252	0.46695
VOC-2007-multilabel (zero-shot)			
mAP	0.7615	0.8579	0.8626
Zero-Shot Retrieval (MSCOCO captions)			
Image Retrieval Recall@5	0.2196	0.3296	0.3385
Text Retrieval Recall@5	0.3032	0.4396	0.4745
Zero-Shot Retrieval (xm3600, English)			
Image Retrieval Recall@5	0.30593	0.43338	0.43700
Text Retrieval Recall@5	0.24293	0.38884	0.42324
Flickr-8K Cross-Modal Metrics			
Euclidean Gap $\downarrow$	0.8276	0.8657	0.8182
JSD $\downarrow$	0.5200	0.2863	0.1455
Wasserstein Distance $\downarrow$	0.4084	0.4166	0.3889
Img-Text Cos Sim (mean) $\uparrow$	0.2723	0.3077	0.3300
Img-Text Cos Sim (std)	0.0362	0.0645	0.0690
Text-Text Cos Sim (mean)	0.6807	0.7243	0.7189
Text-Text Cos Sim (std)	0.1344	0.1377	0.1387

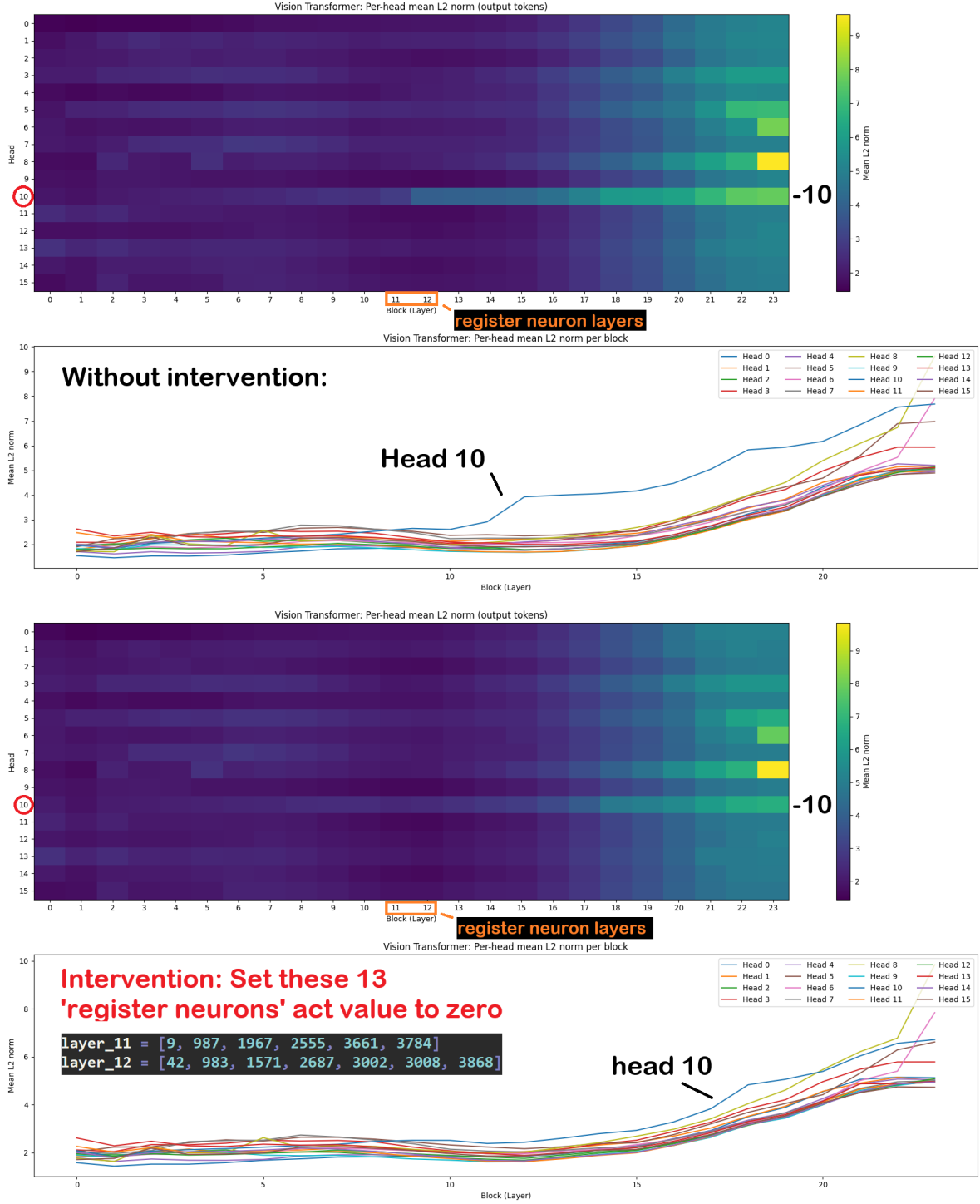


Figure 1: Mean per-head output L2 norm across transformer layers in pre-trained CLIP ViT-L/14, before (top) and after (bottom) ablation of the 13 register neurons. Head 10 shows a sharp reduction in norm after ablation, aligning with other heads.

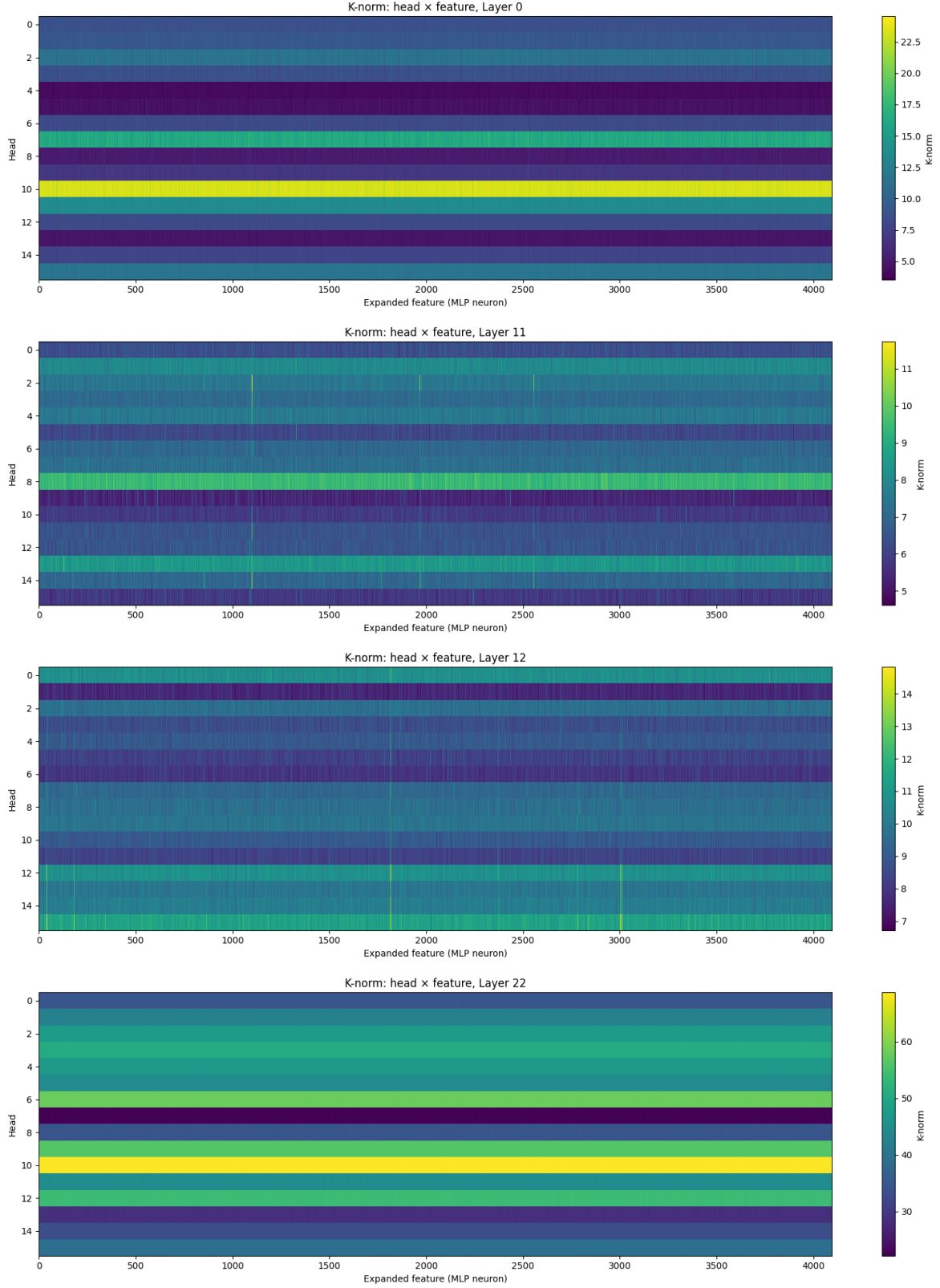


Figure 2: Key norm per attention head in input and penultimate - but not intermediate - layers highlight the role of Head 10; it consistently exhibits the highest key norm (outlier) among all heads at input/penultimate layers.

pre-trained ViT-L/14

"text salience head"

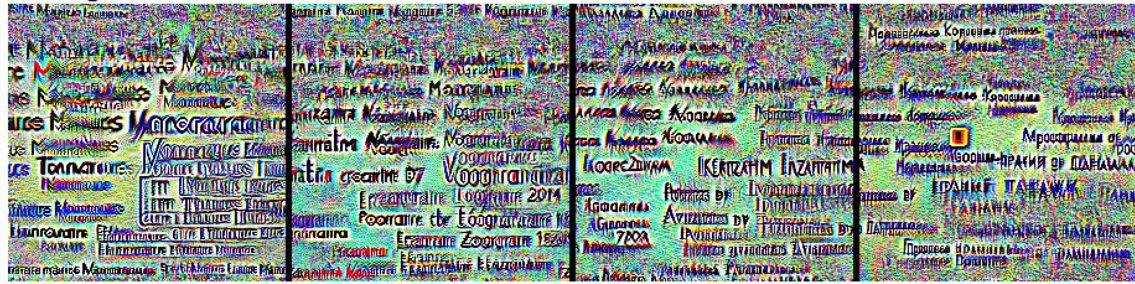
Attention Head 7, normal

Layer 14

15

16

17



Attention Head 7, ablation: 13 register neurons L11, L12



Figure 3: Visualization of feature activation maximization for individual attention heads. Shown: Head 7 (“text salience head”), for which maximization yields images with prominent textual features.

SCAM: A Real-World Typographic Robustness Evaluation for Multimodal Foundation Models

Model	Training data	Accuracy (%)	
		NoSCAM	SCAM
RN50	openai	97.76	36.61 ↓61.15
ViT-B-32	laion2b	98.45	74.68 ↓23.77
ViT-B-16	laion2b	98.71	69.16 ↓29.55
ViT-B-16-SigLIP	webli	99.22	81.40 ↓17.82 #1
ViT-L-14	commonpool_xl	99.48	74.68 ↓24.80
	openai	99.14	40.14 ↓59.00
ViT-L-14-336	openai	99.22	33.85 ↓66.37
ViT-g-14	laion2b	99.05	61.93 ↓37.12
ViT-bigG-14	laion2b	99.40	70.89 ↓28.51

===== BLISS / SCAM TYPOGRAPHIC ATTACK =====			
Zero-shot Acc	Pre-Trained	KO-CLIP	KO-LITE
NoSCAM:	0.9905	0.9897	0.9897
SCAM:	0.4165	#2 0.7823	0.7367
SynthSCAM:	0.3219	0.7358	0.6790

Figure 4: Typographic attack vulnerability on the SCAM benchmark. Pre-trained model exhibits frequent misclassifications under typographic attack, while KO-CLIP (fine-tuned) shows markedly improved robustness, as quantified by accuracy before and after fine-tuning.

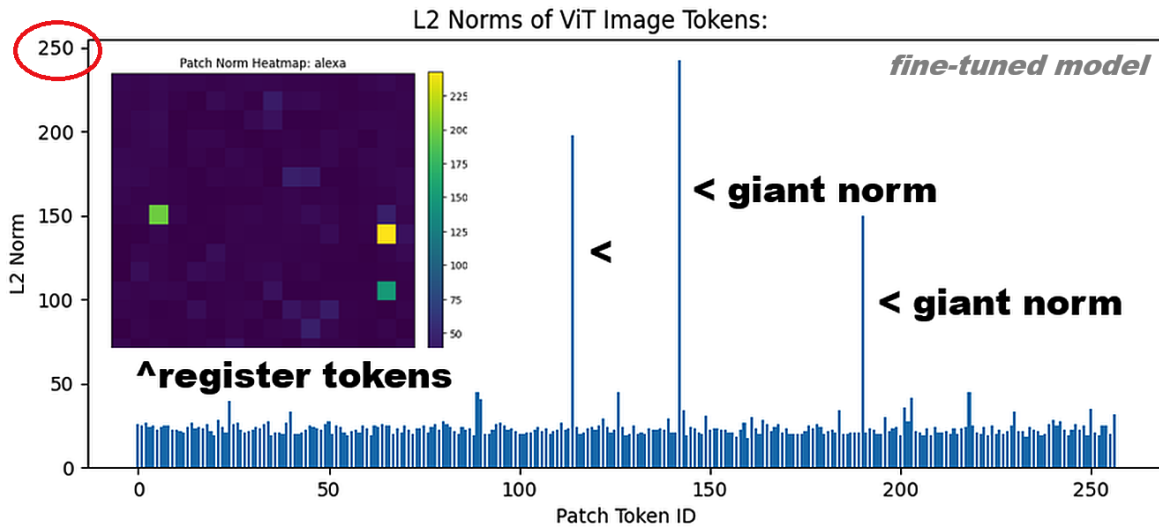
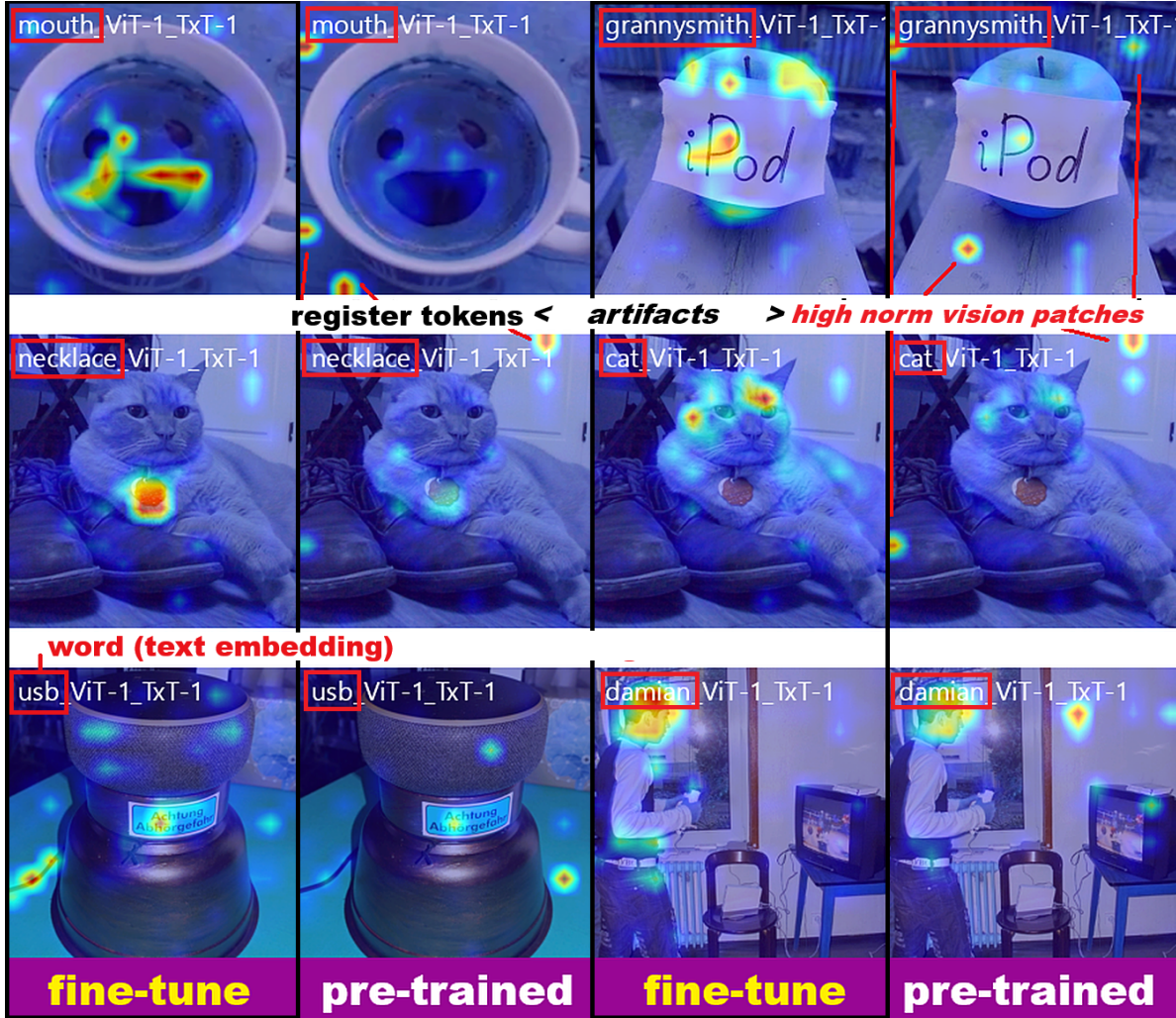


Figure 5: Comparison of attention heatmaps and per-patch L2 norms before and after fine-tuning. Register token artifacts are mitigated in KO-CLIP attention heatmaps, resulting in interpretable maps despite the persistence of high-norm patch tokens ('register tokens') in intermediate to late layers.

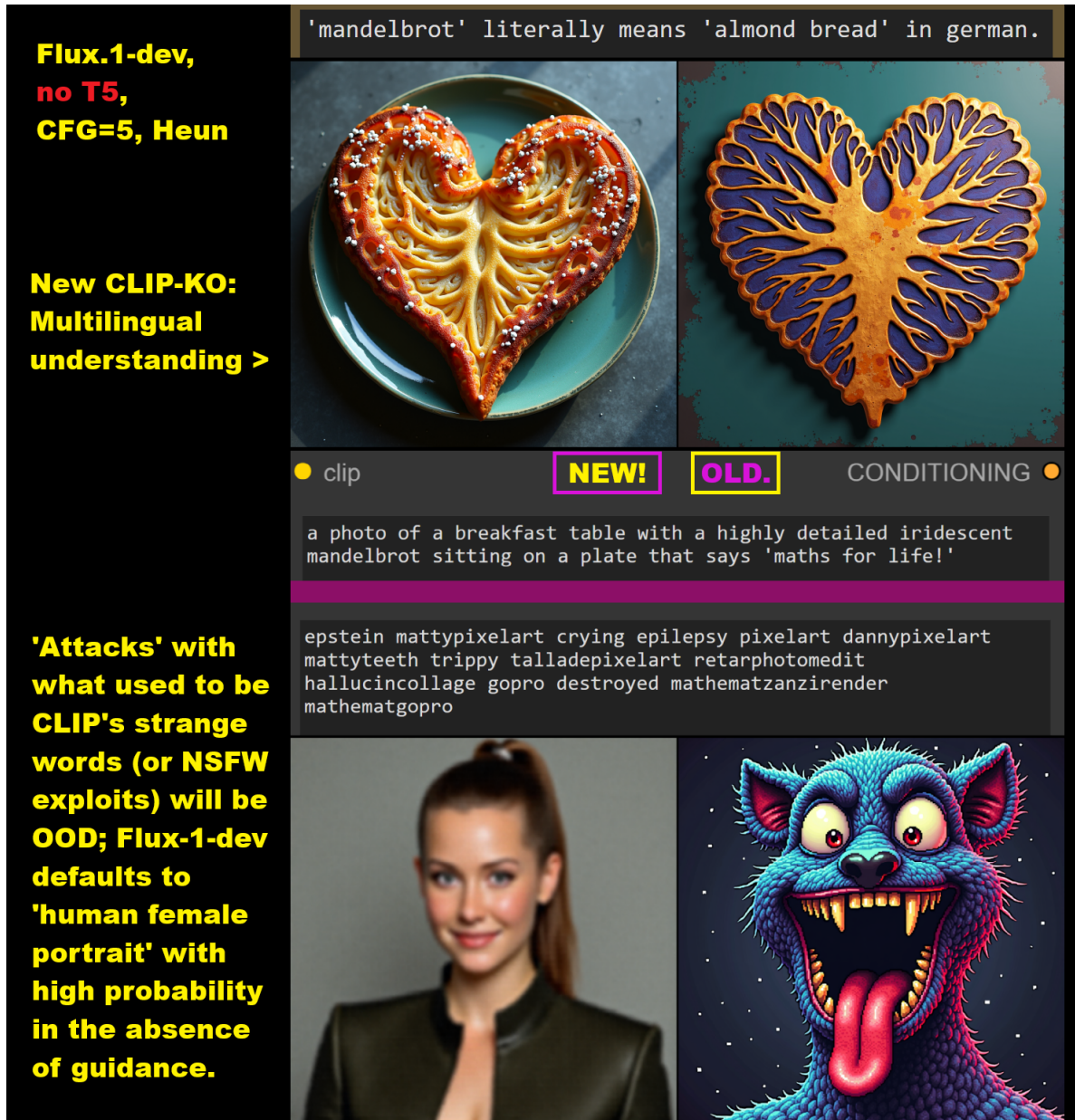


Figure 6: Demonstration of mitigation for prompt injection in text-to-image guidance. With min-cos-sim adversarial training, the 'CLIP-word' prompt produces conceptually aligned images (e.g., a pixel art 'monster'). The KO-CLIP max-cos-sim fine-tuned encoder instead results in an OOD embedding, leading Flux-1.dev to generate a default fallback consistent with “no guidance” behavior (zeroing encoder inputs). Note: No T5 guidance used.